Revisiting Drug Recommendation From a Causal Perspective

Junjie Zhang, Xuan Zang, Hao Chen, Xiaowei Yan, and Buzhou Tang, *Member, IEEE*

***Abstract*—Drug recommendation that aims to provide a prescription for a patient is an essential task in healthcare. Drug molecular graphs provide valuable support for drug recommendation. Existing methods tend to overlook drugs' molecular graphs or use the core substructures of molecular graphs with a rule-based segmentation strategy. However, such methods have several limitations: (1) The rule-based segmentation strategy is inflexible and sub-optimal for extremely complex scenarios. (2) The core substructures derived only consider the drug's chemical characteristics and ignore the patient's health condition. (3) The spurious correlation brought by trivial substructures is disregarded. To address these limitations, we design a novel drug recommendation method from a causal perspective, where a conditional causal representation learner for drug recommendation is proposed. Specifically, we first separate the drug molecular representation into causal and spurious parts depending on various patients' health conditions. Then, we eliminate the spurious correlation caused by the spurious part with causal intervention. Extensive experiments on the MIMIC-III and MIMIC-IV datasets demonstrate that our approach achieves new state-of-the-art performance (e.g., 6.68% Jaccard improvements on MIMIC-III with p-value $\ll$ 0.05).**

***Index Terms*—Causal substructure, drug recommendation, electronic health records (EHRs), molecular graphs.**

I. Introduction

DRUG recommendation that aims to provide a prescription for a patient is an essential task in healthcare. With the integration of computer technology in medical institutions, there is a significant increase in electronic health records (EHRs) and biomedical data, such as drug molecular graphs. These healthcare data provide powerful support for drug recommendation [1], [2]. Drug molecular graphs can help provide more precise personalized treatments, improving efficacy and reducing drug-drug interactions (DDI). However, there are still challenges in fully and rationally applying drug molecular graphs to drug recommendations.

Several deep-learning methods have been developed for drug recommendation to ensure both efficacy and safety. Earlier works [3], [4] recommend drugs solely based on the patient's visit record without considering the drug's molecular properties. Subsequent works [5], [6], [7], [8] incorporate the drug molecular graph representation to fill this gap. However, it's well-known that the chemical properties of molecules depend on part of the core substructure. Therefore, some works [5], [8] aim to model the core substructures to enhance accuracy and safety in drug recommendations. Despite their initial success, existing works still have limitations in the following aspects: (1) Existing works segment the drug molecule into core substructures and trivial substructures with rule-based segmentation strategies (e.g., BRICS [9]), which are inflexible and sub-optimal for extremely complex scenarios. (2) The core substructure of a drug molecular graph is determined by itself and the patient's health condition. Therefore, considering only molecular properties is insufficient. Since the health condition of patients may influence the activity or metabolism of drug molecules under specific circumstances. When obtaining the core substructures of drugs through segmentation, it's essential to consider not only the inherent properties of the drug molecule but also the patient's health condition. (3) After segmentation, the trivial substructure of the drug molecule will introduce noise and lead to spurious correlations. Most existing works focus on the core substructure and neglect the negative effects caused by the trivial substructure.

Based on the aforementioned observations and limitations, we revisit the drug recommendation task from a causal perspective, treating the core substructure as the causal part and the trivial substructure as the spurious part. We begin by constructing a structural causal model (SCM) [10] as illustrated in Fig. 1 and detailed in Section IV-A. With the causal assumption, the causal part and the spurious part are distinguished depending on both the drug's molecular graph and the patient's health condition. Moreover, the spurious part is identified as a confounder [10], which opens backdoor paths between the causal part and the label, creating spurious correlations. To ensure the model accurately learns the causal relationships between variables, it is essential to eliminate the confounding effect.

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Source codes are available at https://github.com/junjzhang7/CRec.

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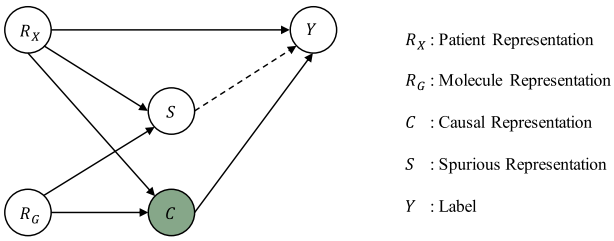


Fig. 1. Structural causal model for the drug recommendation task.

Toward this end, we propose a novel conditional causal representation learner for drug recommendation (CRec), which recommends drugs with the causal representation depending on various patients' health conditions. Specifically, we first employ universal feature extractors to derive the drug molecular and patient representations. Next, we introduce a separation mask to disentangle the molecular representation into causal and spurious parts. This mask is generated considering both the drug molecular and patient representations. Subsequently, we eliminate the spurious correlation caused by the spurious parts with causal intervention. Finally, we make predictions using the intervened molecular and patient representations. By doing so, CRec learns the causality between the causal part and the target while eliminating spurious correlations. Our main contributions can be summarized as follows:

- We revisit the drug recommendation task from a causal perspective and analyze its conditional causal mechanism.
- We introduce a novel conditional causal representation learner for drug recommendation, which recognizes the causal part of a drug molecule depending on various patients' health conditions. Further, we eliminate spurious correlations with causal intervention.
- We conduct comprehensive experiments on the MIMIC-III and MIMIC-IV datasets, and the results demonstrate the effectiveness of the proposed CRec.

II. Related Works

A. Drug Recommendation.

Existing works in the field of drug recommendation can be divided into two categories based on whether or not molecular information is used. Earlier works focused solely on patient information, representing drugs as one-hot vectors. Among them, DMNC [11] and AMANet [12] captured the patient's clinical features and temporal relationships with external memories. However, these methods overlook drug relationships, potentially limiting safety. To address this issue, GAMENet [13] integrated the DDI knowledge graph with a memory module to enhance safety. Similarly, SHAPE [14] incorporated the DDI knowledge graph and utilized curriculum learning to enhance drug recommendation performance. More recent studies [5], [6], [7], [8], [15] have integrated drug molecular graphs into their approaches. SafeDrug [5] and MoleRec [8] considered drug molecule substructures and extracted hierarchical molecular representations. These works demonstrate that incorporating the substructure of drug molecules improves efficacy and safety. However, rule-based segmentation strategies may be inflexible and sub-optimal for extremely complex real-world scenarios. Furthermore, these methods focus on learning statistical correlations between substructures and labels rather than causality, potentially leading to spurious correlations. In this study, we aim to discover the causal representation of drug molecules and alleviate spurious correlations through causal intervention.

B. Causal Inference.

Causal inference is an effective method for identifying causal relationships among multiple variables [16]. It has been widely adopted across various domains of machine learning, including computer vision [17], [18] and graph representation learning [19], [20], [21], [22]. For instance, DisC [22], CMRL [23], and CAL [19] have been developed to disentangle the causal and bias subgraph for graph classification. OrphicX [21] generated causal interpretation for graph neural networks. In the field of drug recommendation, DrugRec [24] was the first to introduce causal inference and apply the front-door adjustment to deconfound recommendation bias. CAMeR [25] applied counterfactual analysis to mitigate spurious correlations in historical visit information. DGMed [26] introduced a bias correction method based on causal inference, which addresses biases among medical entities. However, these approaches do not integrate the drug molecule into their proposed SCM. In this study, we investigate the causal relationships between the patient's health condition and the drug molecule, aiming to enhance drug recommendation by leveraging causal inference.

III. Problem Formulation

Notations: Let $G = \{\mathcal{V}, \mathcal{E}\}$ denotes a molecular graph, where $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ is the atom set, N denotes the number of atoms in the molecule, and $\mathcal{E}$ is the bond set. $\mathbf{X} = [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(t-1)}]$ represents the patient's visit history. For t-th visit of a patient, $\mathbf{x}^{(t)} = [\mathbf{d}^{(t)}, \mathbf{p}^{(t)}, \mathbf{m}^{(t)}]$, where $\mathbf{d}^{(t)} \in \{0,1\}^{|\mathcal{D}|}, \mathbf{p}^{(t)} \in \{0,1\}^{|\mathcal{P}|}, \mathbf{m}^{(t)} \in \{0,1\}^{|\mathcal{M}|}$ are multi-hot vectors representing diagnoses, procedures, and drugs, respectively.[1] $\mathcal{D}, \mathcal{P}, \mathcal{M}$ denote the diagnoses, procedures, and drugs code sets, and $|\cdot|$ is the cardinality.

Drug recommendation: We frame the drug recommendation task as a binary classification problem. Specifically, for a given patient X and a set of all available drugs M, our objective is to determine whether each drug $m \in \mathcal{M}$ is appropriate for the current patient. All appropriate drugs form the final recommended drug combination for the patient.

IV. Method

A. Causal View of Drug Recommendation.

Taking a causal perspective on the drug recommendation task, we construct a Structural Causal Model (SCM), illustrated in Fig. 1. It describes the causal relationships among latent

[1]For brevity, we will omit the visit index t in the following sections.

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variables: patient representation R_X, drug molecular representation R_G, the causal part C of R_G, the spurious part S of R_G, and label Y. The solid links denote the causal relationships between the nodes, while the dashed link denotes the spurious correlation. Here's a detailed explanation of the SCM:

- $R_X \rightarrow C \leftarrow R_G$: C represents the causal part of the drug molecular representation R_G, which is influenced by both the drug molecular structure and the corresponding patient's health condition. This relationship reflects the fact that the activity or metabolism of drug molecules can be affected by the specific circumstances of a patient's health condition.
- $R_X \rightarrow S \leftarrow R_G$: S is the spurious part of the drug molecular representation R_G, acting as a confounder and introducing spurious correlations between R_G and label Y. Similar to the causal part C, S is also influenced by the (R_X, R_G) pair.
- $C \rightarrow Y \leftarrow R_X$: In the drug recommendation task, the final prediction is determined by the causal part of the drug molecular representation and the patient's health condition.
- $S \dashrightarrow Y$: The establishment of this relationship is attributed to the fact that certain drugs may have therapeutic benefits for multiple diseases, despite not being specifically designed to treat any one particular disease. During the learning process, this can cause the model to link the non-specific effects of a drug with the therapeutic effects of a disease, including some substructures, resulting in spurious correlations.

Based on the assumed SCM, we can recognize backdoor paths between C and Y, i.e., $C \leftarrow R_G \rightarrow S \rightarrow Y$, $C \leftarrow R_G \rightarrow S \rightarrow R_X \rightarrow Y$, $C \leftarrow R_X \rightarrow S \rightarrow Y$, and $C \leftarrow R_X \rightarrow Y$. These backdoor paths are the source of spurious correlations that mislead the model into learning an unstable relationship between drugs and labels. To alleviate the spurious correlation, instead of directly modeling the confounded distribution $P(Y|C, R_X)$, we learn the drug molecular representation by blocking the backdoor paths in Fig. 1 with *do-calculus* in causal inference [10] and model the intervened distribution $\tilde{P}(Y|C, R_X) = P(Y|do(C), R_X)$ as follows:

$$\begin{aligned} P(Y|do(C), R_X) &= \tilde{P}(Y|C, R_X) \\ &= \sum_{s \in S} \tilde{P}(Y|C, R_X, s)\tilde{P}(s|C, R_X) \\ &= \sum_{s \in S} \tilde{P}(Y|C, R_X, s)\tilde{P}(s|R_X) \\ &= \sum_{s \in S} P(Y|C, R_X, s)P(s|R_X), \end{aligned} \tag{1}$$

where $P(Y|C, R_X, s)$ is the conditional probability given the causal representation C, current patient representation R_X, and the spurious representation s. And $P(s|R_X)$ denotes the conditional probability of the representation s given the current patient. Intuitively, the backdoor adjustment removes the spurious correlation via a two-step strategy: estimating the causal effect in each stratum of the confounder, weighted by the prevalence of the confounder given the patient, and then summing the causal effects across all strata. This ensures that the overall causal effect is not biased towards certain strata.

B. Model Architecture

In this section, we will go over the details of our proposed CRec model. As shown in Fig. 2, CRec consists of four steps: Firstly, we use universal feature extractors to derive the patient representation and drug molecular representation. Subsequently, we fuse the patient representation and molecular representation to generate the separation mask. Next, we eliminate the spurious correlation induced by the spurious part through causal intervention. Finally, we utilize the intervened representation and the patient representation to make predictions.

1) Encoding With Universal Feature Extractors: We first encode the patient data and the drug molecular graph, using learnable feature extractors. For this purpose, we use $\mathbf{h}_\theta(\cdot) : \mathbf{X} \mapsto \mathbf{E}_x$ to encode the patient's diagnosis and procedure, and $\mathbf{h}_\phi(\cdot) : G \mapsto \mathbf{E}_g$ to encode the drug molecular graph. The process can be formulated as follows:

$$\mathbf{E}_x = \mathbf{h}_\theta(\mathbf{X}), \quad \mathbf{E}_g = \mathbf{h}_\phi(G). \tag{2}$$

Here, $\mathbf{E}_x \in \mathbb{R}^d$ and $\mathbf{E}_g \in \mathbb{R}^{N \times d}$ represent the patient's presentation and the drug molecular representation respectively, with d denoting the dimension of the latent space. The encoder $\mathbf{h}_\theta(\cdot)$ can be any sequential neural network, such as RNN or Transformer, while $\mathbf{h}_\phi(\cdot)$ can be any graph neural network, such as GCN [27], GAT [28], or GIN [29]. Following previous work [8], we adopt RNN and GIN for $\mathbf{h}_\theta(\cdot)$ and $\mathbf{h}_\phi(\cdot)$ respectively.

2) Separation Based on Interaction: As depicted in Fig. 1, the causal and spurious parts of the drug molecular representation are determined by both the molecule itself and the patient's health condition. Therefore, we devise a two-step strategy for separation: (1) Generating a separation mask based on the drug molecular representation and the patient representation; (2) Utilizing the mask to separate the original drug molecular representation into the causal and spurious parts.

In the first step, we design a succinct fusion module to fuse the drug molecular representation and the patient representation. Specifically, given the pair of patient representation $\mathbf{E}_x$ and drug molecular representation $\mathbf{E}_g$, we first calculate the relevance score $\mathbf{w} \in \mathbb{R}^N$ between the molecular representation and the patient representation with a Multi-Layer Perceptron (MLP) followed by a sigmoid function. The entries of $\mathbf{w}$ represent the significance of each atom in treating the patient's disease. We consider this process a coarse-grained drug screening process based on the patient's health condition. Then, we derive the weighted drug molecular representation with the element-wise product. Subsequently, we concatenate the patient representation $\mathbf{E}_x$ with the weighted drug molecular representation $\tilde{\mathbf{E}}_g$ to yield the fused representation $\mathbf{E} \in \mathbb{R}^{N \times 2d}$. This process is formulated as follows:

$$\begin{aligned} \mathbf{E} = \text{Fusion}(\mathbf{E}_x, \mathbf{E}_g) = \mathbf{E}_x || (\mathbf{w} \odot \mathbf{E}_g), \\ where \quad \mathbf{w} = \sigma(\text{MLP}(\mathbf{E}_x)), \end{aligned} \tag{3}$$

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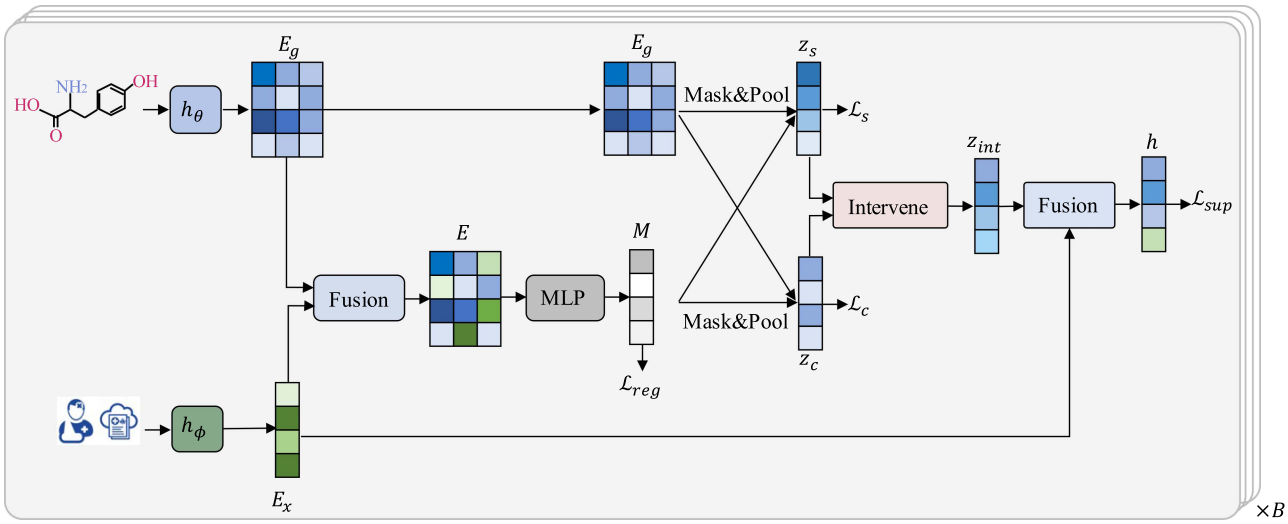


Fig. 2. An overview of CRec. Initially, we encode the input patient data and drug molecular graph to derive their respective representations, denoted as $\mathbf{E}_x$ and $\mathbf{E}_g$. Next, we learn the separation mask based on the fused representation $\mathbf{E}$, which combines the patient and molecular representations. Using the learned mask, we separate the original drug molecular representation into causal and spurious parts, represented as $\mathbf{z}_c$ and $\mathbf{z}_s$, respectively. Following disentanglement, we employ causal intervention to mitigate the spurious correlation introduced by $\mathbf{z}_s$. Finally, we use the intervened representation to predict the label.

where $\sigma(\cdot)$ represents the sigmoid function, $\odot$ denotes the element-wise product, and $||$ denotes the concatenation operation on the hidden demention.

Having obtained the fused representation $\mathbf{E}$, we assign each node a probability of belonging to the causal part. To achieve this, we employ another MLP to learn the probability p_i of the i-th node as follows:

$$p_i = \text{MLP}(\mathbf{E}_i). \tag{4}$$

In the second step, we use a mask $\mathbf{M} \in \mathbb{R}^N$ to disentangle the original drug molecular representation $\mathbf{E}_g$ into the causal part $\mathbf{E}_c \in \mathbb{R}^{N\times d}$ and the spurious part $\mathbf{E}_s \in \mathbb{R}^{N\times d}$ as follows:

$$\mathbf{E}_c = \mathbf{E}_g \odot \mathbf{M}, \quad \mathbf{E}_s = \mathbf{E}_g \odot (\mathbf{1} - \mathbf{M}), \tag{5}$$

where $\mathbf{M}_i \sim \text{Bernoulli}(p_i)$. By doing so, we expect the model to learn the mask based on the causality relationship between the causal part and the label. However, sampling $\mathbf{M}$ from the Bernoulli distribution is non-differentiable, therefore we resort to the commonly-used derivable Gumbel-Softmax trick [30] as follows:

$$\mathbf{M}_i = \text{Gumbel-Softmax}(p_i), \tag{6}$$

We consider this process as a fine-grained substructure screening process based on causality relationships.

Finally, we derive the causal part representation $\mathbf{z}_c \in \mathbb{R}^d$ and spurious part representation $\mathbf{z}_s \in \mathbb{R}^d$ using the readout operator:

$$\mathbf{z}_c = \text{READOUT}(\mathbf{E}_c), \quad \mathbf{z}_s = \text{READOUT}(\mathbf{E}_s). \tag{7}$$

3) Eliminating Spurious Correlation: As discussed in Section IV-A, after separation, the spurious part can introduce spurious correlations through backdoor paths. Therefore, we implement the backdoor adjustment as a pivotal step in mitigating the spurious correlations. As defined in (1), the backdoor adjustment involves randomly combining the various spurious parts S with the causal part C. However, due to the discrete nature of graph data, directly combining the causal substructure with various spurious substructures to generate the intervened graph is not trivial. Therefore, we implicitly conduct the backdoor adjustment in the latent space. Specifically, given the causal part representation $\mathbf{z}_c$ and the spurious part representation $\mathbf{z}_s$, we derive the intervened representation $\mathbf{z}_{int} \in \mathbb{R}^d$ as follows:

$$\mathbf{z}_{int} = \text{MLP}(\mathbf{z}_c || \mathbf{z}'_s), \tag{8}$$

where $\mathbf{z}'_s$ denotes the batch-shuffled $\mathbf{z}_s$. Through random shuffling, the causal intervention combines various spurious parts with the causal part in the representation space, thereby generating the intervened molecular representation.

4) Prediction: After obtaining the intervened molecular representation, we leverage the fusion strategy in (3) to fuse the patient query $\mathbf{E}_x$ and the intervened molecular representation $\mathbf{z}_{int}$ to derive the fused representation $\mathbf{h} \in \mathbb{R}^{2d}$ as follows:

$$\mathbf{h} = \text{Fusion}(\mathbf{E}_x, \mathbf{z}_{int}). \tag{9}$$

To generate the final prediction $\hat{\mathbf{y}}$, we apply another MLP followed by the sigmoid function $\sigma(\cdot)$ as follows:

$$\hat{\mathbf{y}} = \sigma(\text{MLP}(\mathbf{h})). \tag{10}$$

C. Learning Objectives

The proposed CRec is trained end-to-end with a combination of multiple objectives.

Objective 1: Following previous works, we adopt the Multiple Loss [8] as the supervision loss $\mathcal{L}_{sup}$, which comprises binary cross-entropy loss $\mathcal{L}_{bce}$, multi-label margin loss $\mathcal{L}_{multi}$, and ddi loss $\mathcal{L}_{ddi}$ as follows:

$$\mathcal{L}_{bce} = -\sum_{i=1}^{|\mathcal{M}|} \mathbf{y}_i \log(\hat{\mathbf{y}}) + (1 - \mathbf{y}_i) \log(1 - \hat{\mathbf{y}}), \tag{11}$$

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$$\mathcal{L}_{multi} = \sum_{i,j:\mathbf{y}_i=1,\mathbf{y}_j=0} \frac{\max(0, 1-(\hat{\mathbf{y}}_i - \hat{\mathbf{y}}_j))}{|\mathcal{M}|}, \tag{12}$$

$$\mathcal{L}_{ddi} = \sum_{i=1}^{|\mathcal{M}|}\sum_{j=1}^{|\mathcal{M}|} \mathbf{A}_d \cdot \hat{\mathbf{y}}_i \cdot \hat{\mathbf{y}}_j, \tag{13}$$

$$\mathcal{L}_{sup} = \alpha(\beta\mathcal{L}_{bce} + (1-\beta)\mathcal{L}_{multi}) + (1-\alpha)\mathcal{L}_{ddi} \tag{14}$$

where $\mathbf{A}_d$ is the DDI adjacency matrix, and α, β are hyperparameters to control the weights, consistent with MoleRec [8].

Objective 2: For the causal part $\mathbf{z}_c$, intended to represent the causal information of the drug molecule, we aim to maximize its predictive power for the downstream task. Thus, we define the supervised classification loss as follows:

$$\hat{\mathbf{y}}_c = \Phi_c(\mathbf{z}_c), \quad \mathcal{L}_c = \mathcal{L}_{sup}(\hat{\mathbf{y}}_c, \mathbf{y}). \tag{15}$$

where Φ_c is an MLP classifier. We note that the patient representation $\mathbf{E}_x$ has partial predictive power to the label, as shown in Fig. 1. To eliminate the dependence on $\mathbf{E}_x$, we exclude $\mathbf{E}_x$ as part of the input of Φ_c.

Objective 3: We expect that the spurious part $\mathbf{z}_s$ would capture the trivial information of the drug molecule. Thus, we aim to reduce its predictive power for the downstream task. To achieve this, we aim to push its prediction $\mathbf{y}_s$ towards a uniform distribution $\mathbf{y}_{KL}$ as follows:

$$\hat{\mathbf{y}}_s = \Phi_s(\mathbf{z}_s), \quad \mathcal{L}_s = \mathrm{KL}(\hat{\mathbf{y}}_s, \mathbf{y}_{KL}), \tag{16}$$

where Φ_s is an MLP classifier. For a similar concern in Objective 2, to eliminate the interference caused by the predictive ability of $\mathbf{E}_x$, we only use $\mathbf{z}_s$ as the input of Φ_s.

Objective 4: Inspired by the sparsity prior and information bottleneck theory, we aim to achieve a concise causal part representation and compress information into a compact subset. Therefore, we introduce l_0 regularization to encourage the sparsity of the mask $\mathbf{M}$:

$$\mathcal{L}_{reg} = \sum_{i=1}^{N} \mathbf{M}_i, \quad with \quad \mathbf{M}_i = \begin{cases} 1, & if \quad \mathbf{M}_i \neq 0 \\ 0, & otherwise \end{cases} \tag{17}$$

Because the l_0-norm is hard to optimize in practice, we use l_1-norm for approximation: $\mathcal{L}_{reg} \simeq \sum_{i=1}^{N} ||\mathbf{M}_i||_1$, where $||\cdot||_1$ denotes the l_1-norm.

Finally, we combine the losses with the weighted sum form:

$$\mathcal{L} = \mathcal{L}_{sup} + \lambda_1\mathcal{L}_c + \lambda_2\mathcal{L}_s + \lambda_3\mathcal{L}_{reg}, \tag{18}$$

where $\lambda_1, \lambda_2, \lambda_3$ are hyperparameters.

V. Experiments

In this section, we conduct extensive experiments on EHR datasets to answer the following research questions:

1) *RQ1:* Does our proposed CRec outperform the existing drug recommendation methods?
2) *RQ2:* How does each component contribute to the final performance of the proposed CRec?
3) *RQ3:* Does our proposed CRec recommend drugs based on both the patient's condition and the drug's molecular property, while also identifying the core substructure of the drug molecule?
4) *RQ4:* Is the performance of CRec sensitive to the hyperparameters?

A. Experimental Setup

Datasets: We utilize two real-word EHR datasets: MIMIC-III dataset [31] and MIMIC-IV dataset [32], which are widely used in medical research. In MIMIC-III / IV, the diagnosis and procedure codes are based on the ICD-9 code, and the drug codes are based on the NDC (National Drug Code). To ensure fair and standardized comparisons, we adopt the data processing pipeline from PyHealth [33] in our main study. Additionally, we conduct supplementary experiments following the data processing of MoleRec [8]. The statistical information of the processed dataset is listed in Table III.

Baselines: Using the same data processing pipeline, we compare CRec with the following mainstream baselines: (1) RETAIN [34], GAMENet [13], and MICRON [3] which do not incorporate drug molecular information. (2) MoleRec [8] which utilizes the core substructure of the drug molecule.

Evaluation Metrics: Consistent with previous works [8], [13], we employ four commonly used metrics in drug recommendation: Jaccard, F1-score, PRAUC, and DDI. To eliminate randomness, all experimental results are averaged over 5 trials with different random seeds, and the mean and standard deviation are reported.

Implementation Details: We implement CRec and baselines with PyTorch 1.13.0 on a server with two NVIDIA RTX 3090Ti GPUs. Following previous works, we adopt the gate recurrent unit (GRU) and 4-layer GIN [29] for h_ϕ and h_θ, respectively. And we set the embedding size $d = 64$. Our model is trained using Adam optimizer [35] with a learning rate of 0.001. For the hyper-parameters, we set $\lambda_1 = 0.1$, $\lambda_2 = 0.5$, $\lambda_3 = 0.1$ and $\beta = 0.95$ for all datasets.

B. Results on Effectiveness (RQ1)

We report the results of all models in Tables I and II. Overall, CRec achieves the best performance on most evaluation metrics. Specifically, compared to Molerec, CRec achieves a significant improvement of 6.68% in the Jaccard on the MIMIC-III dataset, with a p-value $\ll 0.05$. Moreover, we have the following observations: (1) CRec and Molerec outperform RETAIN, MICRON, and GAMENet, highlighting that incorporating drug molecular information enhances the model's ability to capture drug properties, resulting in more reliable and effective performance. (2) CRec is more effective than MoleRec. This is because using all the molecular features without discrimination can lead to suboptimal performance. The reason for this is that the causal part of a drug molecule depends on both the drug molecule itself and the patient's condition. MoleRec identifies the core substructure using a rule-based segment strategy, without taking into account the patient's health condition. (3) Additionally, our approach outperforms baselines by addressing spurious correlations induced by the spurious part, which is overlooked by existing methods.

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TABLE I
PERFORMANCE COMPARISON ON MIMIC-III / MIMIC-IV DATASETS

Dataset	Model	Jaccard ↑	F1-score ↑	PRAUC ↑	DDI ↓
MIMIC-III	RETAIN	0.4443 ± 0.0027	0.5989 ± 0.0029	0.7468 ± 0.0015	0.0575 ± 0.0009
	GAMENet	0.4212 ± 0.0040	0.5761 ± 0.0039	0.7103 ± 0.0050	0.0613 ± 0.0020
	MICRON	0.4459 ± 0.0017	0.5999 ± 0.0016	0.7284 ± 0.0043	**0.0574 ± 0.0011**
	MoleRec	0.4518 ± 0.0020	0.6072 ± 0.0019	0.7216 ± 0.0012	0.0644 ± 0.0013
	CRec	**0.4820 ± 0.0025**	**0.6352 ± 0.0024**	**0.7518 ± 0.0012**	0.0628 ± 0.0012
MIMIC-IV	RETAIN	0.4412 ± 0.0010	0.5925 ± 0.0010	0.7415 ± 0.0006	0.0662 ± 0.0009
	GAMENet	0.4264 ± 0.0036	0.5765 ± 0.0034	0.7059 ± 0.0009	**0.0595 ± 0.0007**
	MICRON	0.4524 ± 0.0009	0.6028 ± 0.0010	0.7397 ± 0.0003	0.0607 ± 0.0013
	MoleRec	0.4228 ± 0.0013	0.5815 ± 0.0012	0.6937 ± 0.0009	0.0694 ± 0.0005
	CRec	**0.4747 ± 0.0020**	**0.6279 ± 0.0018**	**0.7495 ± 0.0006**	0.0665 ± 0.0007

Ground-truth DDI rate is 0.0754 / 0.0713. The best is marked with boldface and the second best is with underline. The results are reported with the pipeline from PyHealth.

TABLE II
PERFORMANCE COMPARISON ON MIMIC-III DATASET

Dataset	Model	Jaccard ↑	F1-score ↑	PRAUC ↑	DDI ↓
MIMIC-III	RETAIN	0.4866 ± 0.0034	0.6471 ± 0.0032	0.7593 ± 0.0035	0.0871 ± 0.0013
	DMNC	0.4550 ± 0.0031	0.6160 ± 0.0031	0.6757 ± 0.0029	0.0801 ± 0.0011
	GAMENet	0.5037 ± 0.0015	0.6601 ± 0.0014	0.7673 ± 0.0024	0.0859 ± 0.0005
	SafeDrug	0.5126 ± 0.0028	0.6691 ± 0.0023	0.7655 ± 0.0022	0.0773 ± 0.0006
	4SDrug	0.4800 ± 0.0027	0.6404 ± 0.0024	0.7611 ± 0.0026	0.0703 ± 0.0011
	DrugRec	0.5220 ± 0.0034	0.6771 ± 0.0031	0.7720 ± 0.0036	**0.0597 ± 0.0006**
	MoleRec	0.5305 ± 0.0033	0.6843 ± 0.0029	0.7736 ± 0.0027	0.0724 ± 0.0008
	CRec	**0.5371 ± 0.0005**	**0.6907 ± 0.0003**	**0.7810 ± 0.0004**	0.0680 ± 0.0028

The best is marked with boldface and the second best is with underline. The results are reported with the pipeline from MoleRec as supplementary.

TABLE III
DATA STATISTICS OF PROCESSED MIMIC-III / IV WITH PYHEALTH& MOLEREC PIPELINES

Item	Size	
	MIMIC-III /IV (PyH.)	MIMIC-III (Mol.)
# of patients	5449/43 939	6350
# of visits	14 141/147 406	15 032
avg. of visits	2.5952/3.3548	2.37
# of unique conditions	4491/19 184	1958
avg. of conditions	31.652/48.9885	10.51
# of unique procedures	1412/10 603	1430
avg. of procedures	9.4601/8.763	3.84
# of unique drugs	193/200	131
avg. of drugs	29.7427/23.8324	11.44

C. Ablation Studies of Components (RQ2)

To validate the effectiveness of our proposed components, we design several variants. (1) We remove the separation mechanism to validate its effectiveness. This variant is denoted as "w/o separation". In this variant, we directly predicted the target with $\mathbf{E}_x||\mathbf{E}_g$. (2) To validate the significance of patient features in the separation, we design a variant that produces the mask solely based on the drug molecular presentation, denoted as w/o interaction. (3) To explore the impact of each learning objective, we remove each objective separately from the final loss including w/o $\mathcal{L}_c$, w/o $\mathcal{L}_s$, and w/o $\mathcal{L}_{reg}$.

Table IV reports the results of variants. We have the following observations: (1) The performance substantially declined without the separation mechanism. This indicates the proposed separation mechanism is effective and separating the drug molecular graph into causal and spurious parts is beneficial. (2) The absence of patient features in the separation results in a significant performance decline, suggesting that both the drug molecular representation's causal and spurious parts are influenced by both the molecule itself and the patient's health condition. (3) Removing the objective $\mathcal{L}_c$ or $\mathcal{L}_s$ results performance degradation. This indicates that it is indispensable to constrain the learning of causal and spurious parts. The result without regulation shows that a sparse and compact causal representation benefits representation learning. By comparing the results, we concluded that all the proposed components have a positive impact on the final performance.

D. Qualitative Analysis (RQ3)

To evaluate the effectiveness of CRec, we randomly select two patients from the test dataset. As depicted in Table V, compared with MoleRec, CRec exhibits higher predictive accuracy and fewer unseen predictions. Additionally, in the context where both patients suffer from 2761 (*hyposmolality and/or hyponatremia*), we have some specific observations: (1) CRec recommends B05X[2] for both patients, indicating that CRec captures commonalities between patients and recommends drugs based on their health condition. (2) CRec recommended A12B[3] and A12C[4] for patient A, even though they are not present in patient A's ground truth. However, we consider it reasonable for the following reasons: Firstly, B05X and A12B are two different ATC (Anatomical Therapeutic Chemical Classification System) codes for potassium chloride under different classification standards. Secondly, A12C has similar effects to B05X and can be used as a substitute to treat 2761. This can also be proved by

[2]B05X: The ATC code of Potassium chloride, a potassium salt used to treat hypokalemia.

[3]A12B: Another ATC code of Potassium chloride, assigned under a different classification standard.

[4]A12C: The ATC code of Sodium chloride, an ingredient found in a variety of nutritional products as a source of electrolytes and water.

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TABLE IV
ABLATION STUDIES OF CREC ON MIMIC-III

Method	Jaccard ↑	F1-score ↑	PRAUC ↑	DDI ↓
w/o separation	0.4690 ± 0.0024	0.6268 ± 0.0025	0.7282 ± 0.0036	0.0655 ± 0.0030
w/o interaction	0.4754 ± 0.0060	0.6313 ± 0.0047	0.7454 ± 0.0010	0.0635 ± 0.0020
w/o $\mathcal{L}_s$	0.4748 ± 0.0086	0.6292 ± 0.0072	0.7506 ± 0.0060	0.0643 ± 0.0032
w/o $\mathcal{L}_c$	0.4760 ± 0.0080	0.6333 ± 0.0075	0.7414 ± 0.0087	0.0679 ± 0.0020
w/o $\mathcal{L}_{reg}$	0.4757 ± 0.0030	0.6326 ± 0.0029	0.7440 ± 0.0041	0.0682 ± 0.0035
CRec	**0.4820 ± 0.0025**	**0.6352 ± 0.0024**	**0.7518 ± 0.0012**	**0.0628 ± 0.0012**

The best is marked with boldface.

TABLE V
CASE STUDY ANALYSIS

Diagnosis (ICD9)	Recommended Medication Combination (ATC)
A: 25541, 5849, 42822, 2761, 4254, 78550, 4280, 5853, 3970, 27401, 42731, 25000, 28521, V4502, V5861, V5867	**GT**: A02B, A07D, A10A, B01A, B02B, B05X, C01A, C01B, C01C, C07A, C09C, G01A, H01A, H01B, H02A, H04A, J01C, J01X, M04A, N02A, N02B, N05A, N06A, R01A, R03B, V06D
	MoleRec: correct: B02B, C01A, C01B, C01C, J01C, N02A, N02B unseen: A02A, A03B, A05B, A06A, A07A, A07F, A07X, A08A, A09A, A10B, A11H, A12A, A12B, A16A, B03A, B05A, B05B, B05C, B06A, C01E, C02D, C02L, C03A, C03B, C04A, C05A, C05B, C05C, C08D, C09X, C10A, D05A, D06B, D07A, D07C, D07X, D08A, D10A, D11A, G03F, G03X, H02C, J01A, J01M, N01A, N01B, N05B, N05C, R02A, R03A, R03C, R05F, R06A, S01C, S01K, S02C, S03C, V03A, V04C
	CRec: correct: A02B, A10A,B01A, B05X, C01A, C01B, C01C, C07A, H04A, J01C, J01X, M04A, N02A, N02B, N06A, R03B, V06D unseen: A02A, A04A, A06A, A11C, A12A, A12B, A12C, B03A, C01D, C03B, C03C, C05A, C09A, C10A, D04A, J01D, J01M,N03A, N05B, N05C, R03A,V04C
B: 5770, 5859, 4258, 42731, 2761, 4280, 2875, 40491, 57420, 25040, 2720, V4502, 28521	**GT**: A02A, A02B, A06A, A10A, A12B, A12C, B01A, B02B, B05X, C01A, C01B, C03C, C07A, C09C, M04A, N02A, N02B, V04C
	MoleRec: correct: A02A,A12B, C01A, C03C, N02A, V04C unseen: A03F, A05B, A09A, A12A, A16A, B02A, B05C, B06A, C01C, C03B, C05C, C08D, C09X, J01A, J01C, J01M, M05B, N01B, N05B, N05C, V03A
	CRec: correct: A02A, A02B, A06A, A10A, A12B, A12C, B01A, B02B, B05X, C01A, C03C, C07A, N02A, N02B, V04C unseen: A04A, C01D, C03D, C09A, C10A, N06A, V06D

Here, unseen denotes drugs predicted by the model but not present in the ground truth (GT).

patient B's ground truth, where medications prescribed for 2761 include B05X, A12B, and A12C. This demonstrates that CRec recommends drugs based on both the patient's health condition and the drug's molecular properties, effectively capturing the causal relationship between drugs and diseases.

We also visualize the molecular graphs of A02B,[5] N02B,[6] A02A,[7] and C01C[8] in Fig. 3 and draw the following conclusions: (1) For A02B, the model highlights the methyl group and the triple bond structure between nitrogen atoms. This aligns with the chemical domain knowledge that methyl increases the drug's lipophilicity, making it easier to pass through the gastrointestinal wall. The triple bond between nitrogen atoms enhances the drug's binding affinity with H2 receptors, increasing its pharmacological activity. (2) Regarding N02B, the model focuses on the carbonyl group. This group is hydrophilic, facilitating aspirin's solubility in water. The water solubility of aspirin allows it to dissolve rapidly and be absorbed in the body. (3) The model identifies the magnesium ion and the carbonyl group in A02A as crucial for its mechanism of action. These functional groups can react with H^+ in stomach acid, producing water and carbon dioxide, helping to neutralize stomach acid. (4) For C01C, the model identifies the secondary hydroxyl group as crucial, as it enhances the β-receptor agonist effect of etilefrine, leading to increased cardiac contractility and heart rate, thereby raising

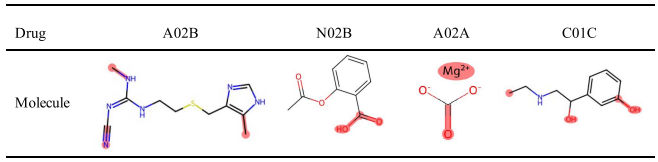


Fig. 3. The visualization of the molecular graphs. (Red highlight: Predicted causal substructure.).

[5]A02B: The ATC code of Cimetidine, a histamine H2 receptor antagonist used to manage GERD, peptic ulcer disease, and indigestion.

[6]N02B: The ATC code of Aspirin, a salicylate used to treat pain, fever, inflammation, migraines, and reduce the risk of major adverse cardiovascular events.

[7]A02A: The ATC code of Magnesium carbonate, an antacid used for symptomatic relief of heartburn, indigestion, and upset stomach.

[8]C01C: The ATC code of Etilefrine, an adrenergic agonist primarily of α1 and β 1 receptors used as an anti-hypotensive.

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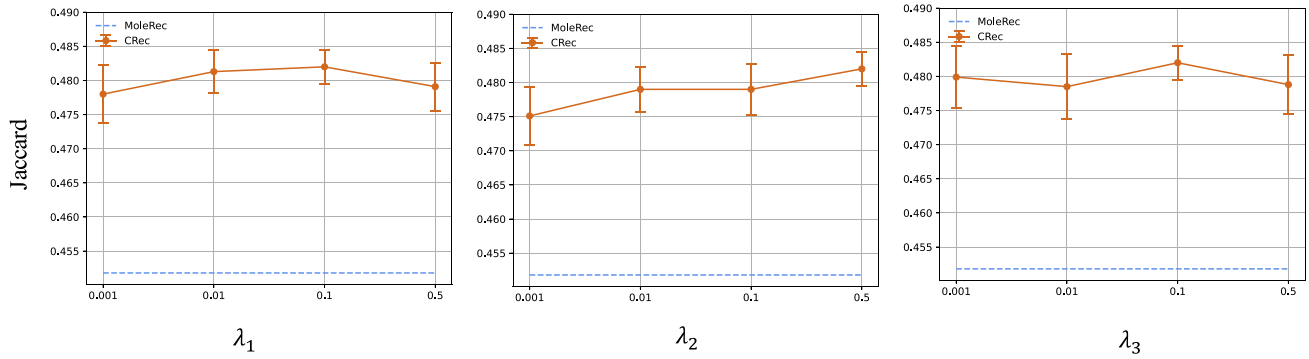


Fig. 4. Hyperparameters sensitivity analysis on the MIMIC-III dataset.

blood pressure. This is beneficial for managing symptoms such as hypotension and heart failure in patients. The visualization analysis demonstrates the explainability of CRec, illustrating its decision-making process based on molecular features.

E. Sensitivity Analysis (RQ4)

We conduct a series of sensitivity analyses about the hyperparameters λ_1, λ_2, and λ_3, which control the strength of the separation and the sparsity of the mask, respectively. In this setting, We fix other parameters when adjusting one parameter. The results are shown in Fig. 4. We have the following observations: (1) Although the performance of our model fluctuates under different parameters, it consistently surpasses the baseline model. (2) CRec performs worse when λ_2 is lower. This is because a smaller KL loss may not completely separate the spurious part. (3) CRec performs best when $\lambda_3 = 0.1$ indicates that an inappropriate sparsity level (or information compression strength) can hinder the model's ability to effectively separate causal and non-causal components.

VI. Conclusion

In this work, we revisit the drug recommendation task from a causal perspective and propose CRec that disentangles the drug molecular representation into causal and spurious parts based on the health conditions of different patients and drug molecular graphs. Additionally, we notice and eliminate the spurious correlation caused by the spurious part. Extensive experiments on the MIMIC-III and MIMIC-IV datasets demonstrate the effectiveness of CRec. While our focus in this work is on disentangling the drug's molecular representations, we recognize the potential for spurious correlations induced by the patient's health condition. We also acknowledge that our current model has limitations in addressing the cold start problem. These areas represent possible future directions for further research.

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